

CHAPTER 9

HISTOGRAM PROCESSING

WHAT WILL WE LEARN?

- What is the histogram of an image?
- How can the histogram of an image be computed?
- How much information does the histogram provide about the image?
- What is histogram equalization and what happens to an image whose histogram is equalized?
- How can the histogram be modified through direct histogram specification and what happens to an image when we do it?
- What other histogram modification techniques can be applied to digital images and what is the result of applying such techniques?

What is a histogram?

- The histogram of a monochrome image is a graphical representation of the frequency of occurrence of each gray level in the image.
- The data structure that stores the frequency values is a 1D array of numerical values, h , *whose individual elements store the number (or percentage) of image pixels that correspond to each possible gray level.*

What is a histogram?

$$h(k) = n_k = \text{card}\{(x, y) | f(x, y) = k\}$$

where:

$k = 0, 1, \dots, L - 1$, where L is the number of gray levels of the digitized image; and $\text{card}\{\dots\}$ denotes the cardinality of a set, i.e. the number of elements in that set (n_k).

A normalized histogram can be mathematically defined as:

$$p(r_k) = \frac{n_k}{n}$$

where:

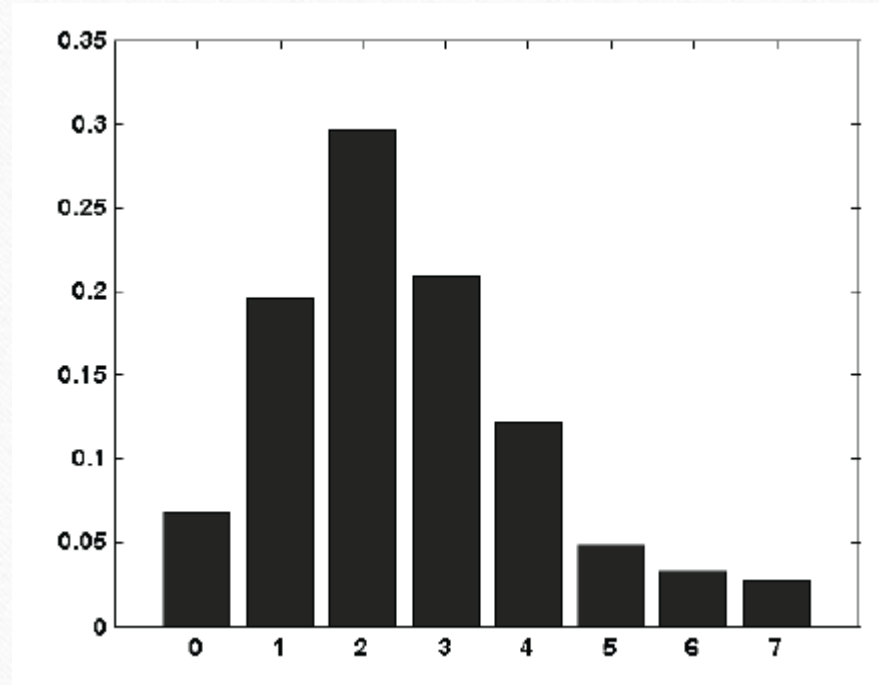
n = total number of pixels in the image; and

$p(r_k)$ = probability (percentage) of the k -th gray level (r_k).

Histogram example

- Example 9.1: histogram for a hypothetical image containing 128×128 pixels and 8 gray levels.

Gray level (r_k)	n_k	$p(r_k)$
0	1120	0.068
1	3214	0.196
2	4850	0.296
3	3425	0.209
4	1995	0.122
5	784	0.048
6	541	0.033
7	455	0.028
Total	16384	1.000

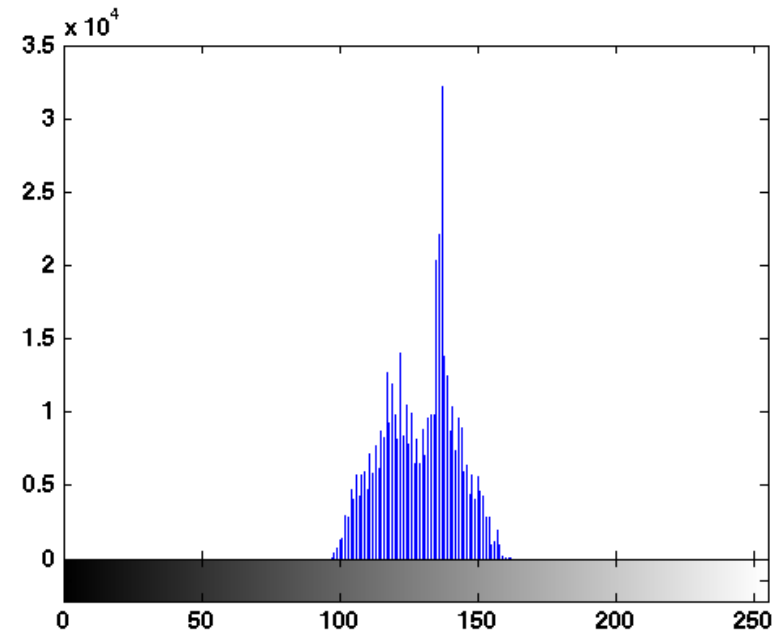


Examples of images and their histograms

(Example 9.2)



- In MATLAB: `imhist`



Interpreting image histograms

- Histograms have become a popular tool for conveying image statistics and helping determine certain problems in an image.
- A histogram carries significant qualitative and quantitative information about the corresponding image (e.g., minimum, average, and maximum gray level values, dominance of bright or dark pixels, etc.).
- A histogram is not enough to draw qualitative conclusions about the overall quality of the image, presence or absence of noise, etc.

Interpreting image histograms

- Although a histogram provides the frequency distribution of gray levels in an image, it tells us nothing about the spatial distribution of the pixels whose gray levels are represented in the histogram.
- Histograms can be used whenever a statistical representation of the gray level distribution in an image is desired.
- Histograms can also be used to enhance or modify the characteristics of an image, particularly its contrast.

Histogram equalization

- Example 9.3:

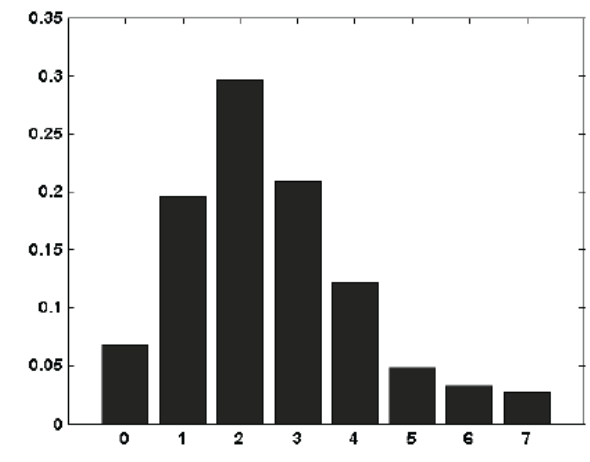
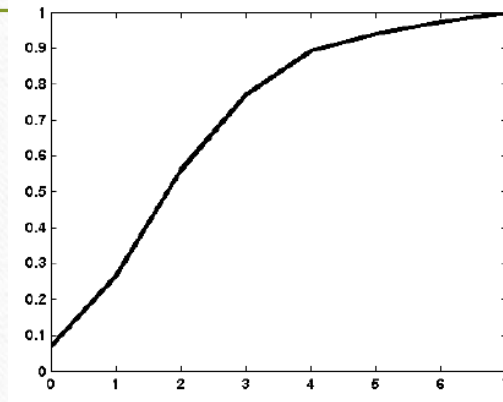
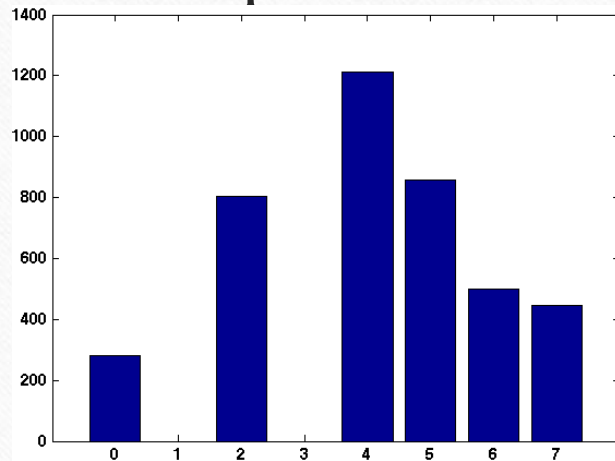


TABLE 9.1 Example of a Histogram

Gray Level (r_k)	n_k	$p(r_k)$
0	1120	0.068
1	3214	0.196
2	4850	0.296
3	3425	0.209
4	1995	0.122
5	784	0.048
6	541	0.033
7	455	0.028
Total	16,384	1.000

TABLE 9.2 Equalized Histogram: Values

Gray Level (s_k)	n_k	$p(s_k)$
0	1120	0.068
1	0	0.000
2	3214	0.196
3	0	0.000
4	4850	0.296
5	3425	0.209
6	1995	0.122
7	1780	0.109
Total	16,384	1.000

Example:

- Calculate and equalize histogram using the cdf of the image with $\text{new_level} = 4$

$$I = \begin{pmatrix} 1 & 2 & 4 & 6 & 7 \\ 2 & 1 & 3 & 4 & 5 \\ 7 & 2 & 6 & 9 & 1 \\ 4 & 1 & 2 & 1 & 2 \end{pmatrix}$$

Solution

g	h(g)	t(g)	f(g)
1	5	5	0
2	5	10	1
3	1	11	1
4	3	14	2
5	1	15	2
6	2	17	2
7	2	19	3
9	1	20	3

$$I_{kq} = \begin{pmatrix} 0 & 1 & 2 & 2 & 3 \\ 1 & 0 & 1 & 2 & 2 \\ 3 & 1 & 2 & 3 & 0 \\ 2 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Histogram equalization

- In MATLAB: `histeq` Example 9.4:

```
I = imread('sydney_low_contrast.png');
```

```
I = im2double(I);
```

```
J = histeq(I);
```

```
figure, subplot(2,2,1), imshow(I), ...
```

```
    subplot(2,2,2), imshow(J), ...
```

```
    subplot(2,2,3), imhist(I), ylim('auto'), ...
```

```
    subplot(2,2,4), imhist(J), ylim('auto')
```

Global vs. local histogram equalization

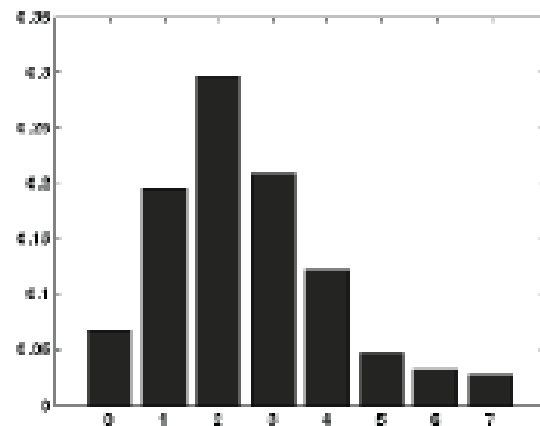
- In MATLAB: `histeq` and `adapthisteq`

```
I = imread('coins.png');  
figure, subplot(1,2,1), imshow(I), ...  
        subplot(1,2,2), imhist(I), ylim('auto')  
  
J = histeq(I);  
figure, subplot(1,2,1), imshow(J), ...  
        subplot(1,2,2), imhist(J), ylim('auto')  
  
K = adapthisteq(I);  
figure, subplot(1,2,1), imshow(K), ...  
        subplot(1,2,2), imhist(K), ylim('auto')
```

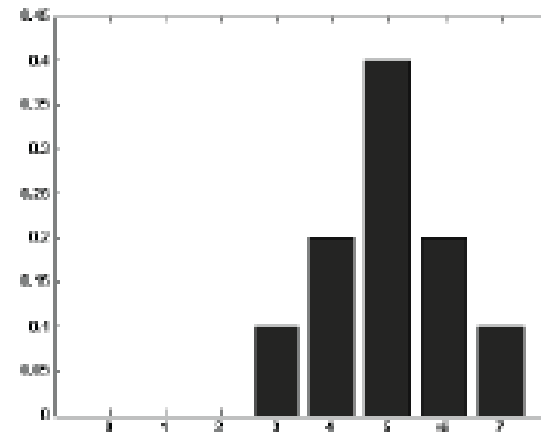
Direct histogram specification

- Example 9.6:

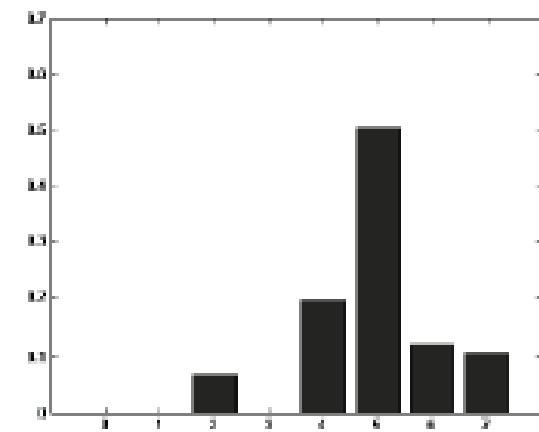
Original



Desired



Result



Direct histogram specification

- In MATLAB: `histeq`

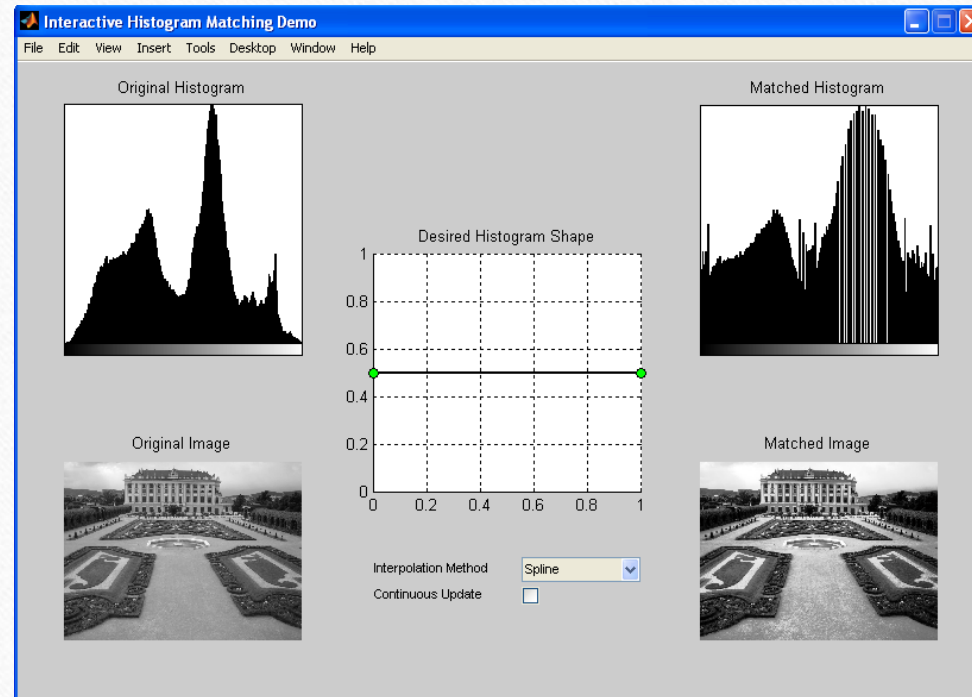
```
I = imread('sydney_low_contrast.png');
Id = im2double(I);
figure, imhist(Id), ylim('auto'), ...
    title ('Original histogram');

des_hist = uint8(zeros(1,256));
des_hist(1:128) = linspace(256,0,128);
des_hist(129:end) = linspace(0,256,128);
x_axis = 0:255;
figure, bar(x_axis, des_hist), axis tight,
...
    title('Desired histogram');

hgram = im2double(des_hist);
Jd = histeq(Id,hgram);
figure, imhist(Jd), ylim('auto'), ...
    title ('Resulting histogram');
```

Direct histogram specification

- Interactive histogram matching tool (ihmdemo)



Other histogram modification techniques

- Histogram sliding
(Example 9.8):



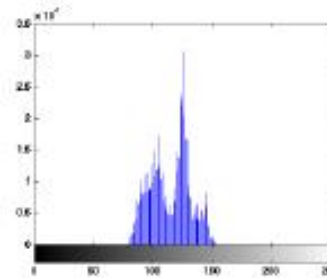
(a)



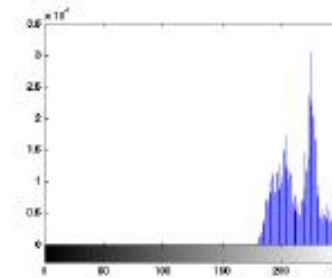
(b)



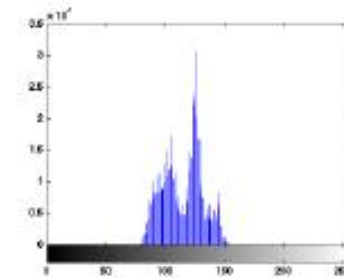
(c)



(d)



(e)



(f)

Histogram sliding

- In
MATLAB:

**imadd and
imsubtract**

```
I = imread('schonbrunn_gray_low_contrast.png');  
figure, imhist(I), ylim('auto'), title ('Original histogram');  
  
I2 = imadd(I, 50);  
figure, imhist(I2), ylim('auto'), ...  
    title ('Sliding to the right by 50');  
  
I3 = imsubtract(I,50);  
figure, imhist(I3), ylim('auto'), ...  
    title ('Sliding to the left by 50');
```

Histogram stretching (Example 9.9):

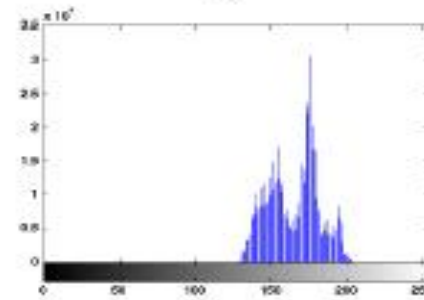
$$s = \frac{r - r_{min}}{r_{max} - r_{min}} \cdot (L - 1)$$



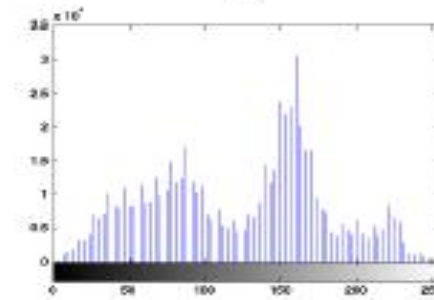
(a)



(b)



(c)



(d)

Histogram shrinking (Example 9.10):

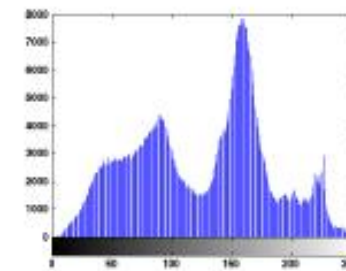
$$s = \left[\frac{s_{max} - s_{min}}{r_{max} - r_{min}} \right] (r - r_{min}) + s_{min}$$



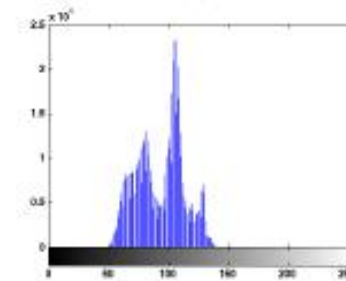
(a)



(b)



(c)



(d)

Histogram stretching and shrinking

- In MATLAB:
- **imadjust**

```
%% Histogram stretching
I = imread('schonbrunn_gray_low_contrast.png');
figure, imhist(I), ylim('auto'), title ('Original histogram');
I2 = imadjust(I);
figure, imhist(I2), ylim('auto'), title ('After histogram stretching');
figure, subplot(1,2,1), imshow(I), subplot(1,2,2), imshow(I2)

%% Histogram shrinking
I = imread('schonbrunn_gray.png');
figure, imhist(I), ylim('auto'), title ('Original histogram');
Id = im2double(I);
Jd = imadjust(Id, [], [49/255 140/255]);
J = uint8(255.*Jd);
figure, imhist(J), ylim('auto'), title ('After histogram shrinking');
figure, subplot(1,2,1), imshow(I), subplot(1,2,2), imshow(J)
```

Hands-on

- Tutorial 9.1: Image histograms (page 188)
- Tutorial 9.2: Histogram equalization and specification (page 191)
- Tutorial 9.3: Other histogram modification techniques (page 195)